

Two equations locate one shared solution.

Lesson topic: solving simultaneous linear equations by linking elimination to a line intersection.

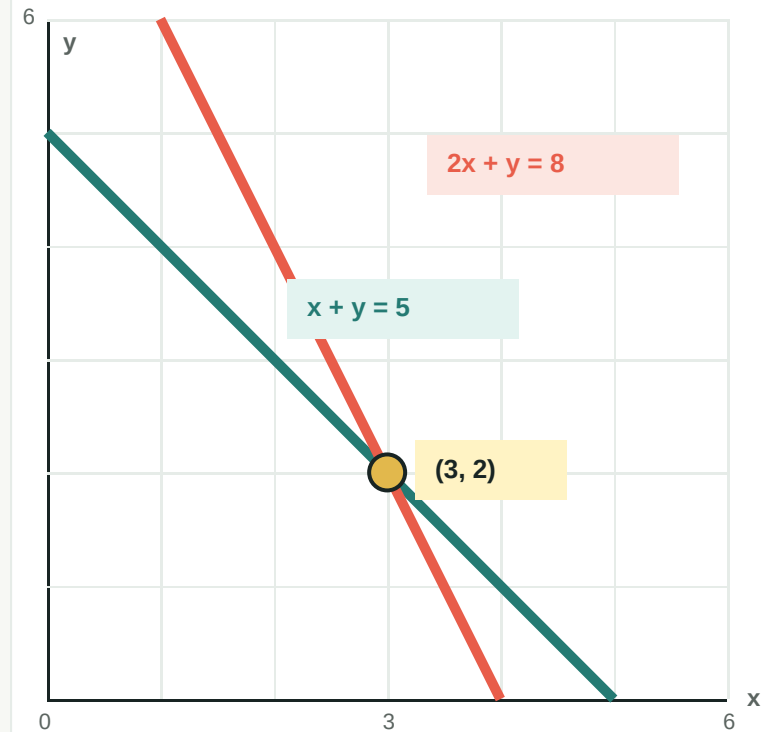
Starter example

$$\begin{aligned}x + y &= 5 \\ 2x + y &= 8\end{aligned}$$

A solution must satisfy both equations.

- Subtract the first equation.
- $x = 3$
- Substitute: $3 + y = 5$
- $y = 2$, so the solution is $(3, 2)$.

Graphical meaning



Knowledge target

Pupils connect the algebraic answer to the intersection of two straight lines.

The hour moves from meaning to method to independent use.

0-8 min Retrieve Rearrange to $y = mx + c$; plot two points; solve one-step equations.	8-18 min Hook Which point works in both rules? Pupils test points on mini-whiteboards.	18-33 min Model Plot both lines, then eliminate the shared y term to solve exactly.	33-48 min Practise Pairs match graph, equations, and solution before independent practice.	48-56 min Check Hinge question: choose the solution and explain a common error.	56-60 min Exit One short problem plus: 'How would this look on a graph?'
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Why this order

Retrieval secures prerequisites, the graph gives meaning before procedure, and the hinge question catches misconceptions before pupils leave.

Engagement comes from pupils making and testing the constraints.

One adult + one child ticket costs GBP 5. Two adults + one child costs GBP 8. What is each ticket price?

1 See it

Test points against both equations, then locate the shared point on a graph.

Resource: printed axes or Desmos/GeoGebra

2 Say it

Pairs explain why a point can satisfy one equation but still fail the system.

Resource: sentence stem and mini-whiteboards

3 Do it

Use elimination to solve, then interpret x and y in the ticket context.

Resource: equation cards and exit slip

Point	$x + y = 5$	$2x + y = 8$
(2, 3)	yes	no
(3, 2)	yes	yes
(4, 1)	yes	no

Pupils see the same idea three ways: table, graph, and equations.

That makes the later symbolic method less arbitrary.

I would check whether pupils know why the method works, not just the answer.

Hinge question

Which point solves both equations?

$$x + y = 5$$

$$2x + y = 8$$

A (2, 3)

B (3, 2)

C (4, 1)

D (5, 0)

Diagnostic use

A, C, and D satisfy the first equation only. B satisfies both, so it is the solution.

Resources and adaptation

- Mini-whiteboards for whole-class checks.
- Projected graph or dynamic graphing tool.
- Printed axes and matching cards for pairs.
- Scaffold: same y coefficient first; colour-code subtraction.
- Challenge: parallel lines, no solution, or linear/quadratic extension.

Exit ticket

Solve $p + q = 9$ and $p - q = 1$, then explain where the answer appears on a graph.

Success: define solution, eliminate accurately, interpret the intersection.